

Samuele TONDELLI

Computer Science and Engineering Student

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“ SUMMARY

Final-year master’s student in Computer Science and Engineering focused on ML systems, GPU computing, and performance engineering. Currently optimizing GPU inference of a graph transformer for protein docking. Background in low-level systems programming, FPGA acceleration, and embedded software. Seeking **internship and research opportunities** between **11/26 - 07/27**.

 EDUCATION

2024 - Present	Master in Computer Science and Engineering, Politecnico di Milano , expected to graduate in July 2027
2021 - 2024	Bachelor in Computer Engineering, University of Modena and Reggio Emilia . Thesis : "Optimization and synthesis on FPGA of the logarithm of softmax function for neuromorphic accelerators". 110 cum laude

 SKILLS

Languages	C, C++, Python, Rust, Assembly, Go, Java
GPU & performance	CUDA, SIMD, Triton, AVX-512, TensorRT, NSight Sytems/Compute
ML frameworks	PyTorch, Scikit-learn, Hydra, TensorFlow
Hardware & Embedded	Vitis HLS, STM32, Verilog, Xilinx Zynq FPGAs, Raspberry Pi
Dev tools	Linux, Git, Make, CMake, GCC, GDB, Valgrind, Vivado, ModelSim

 EXPERIENCE

Feb 2026 Sep 2025	Student Researcher Genomic ML, NECSTLAB, POLITECNICO DI MILANO, Italy ➤ Conducting survey study on ML approaches for single nucleotide variant classification. ➤ Benchmarked 4 models (MLP, XGBoost, Logistic Reg, Random Forest) across 144 genomic datasets. ➤ Reduced training resource usage up to 10x while minimally impacting test performance. ➤ Analyzed pathogenicity on variants of uncertain significance (90% consistency across 72 models). ➤ Paper accepted at IEEE EMBC 2026 (Paper No. 3391). ML Genomics Python Hydra Scikit-learn
Sep 2025 Mar 2025	Student Researcher FPGA Deep Space, NECSTLAB, POLITECNICO DI MILANO, Italy ➤ Accelerated ECOS library for autonomous navigation of deep space satellites. ➤ Developed accelerator for sparse matrix computations (SpMV and LDLt) using Vitis HLS and Vivado. ➤ Deployed IP on Kria KD240 and developed a C Linux driver, achieving 2x speedup over software. ➤ Completed in the 2025 edition of the AMD Open Hardware competition. FPGA HLS Vitis Vivado Xilinx Embedded systems C
May 2024 Mar 2023	Software Engineer Intern Formula SAE, MORE MODENA RACING, UNIMORE, Italy ➤ Co-led a team of 4 developing software for combustion and hybrid Formula SAE cars. ➤ Developed live, wireless telemetry (1 Mbps over 500m) using Raspberry Pi 5 and Grafana. ➤ Developed car dashboard software on STM32 controller reading CAN bus data. Automotive C Raspberry Pi Live telemetry Grafana STM32 CAN

 PROJECTS

GPU ACCELERATED GRAPH TRANSFORMER INFERENCE Master’s thesis Politecnico di Milano, private repository Optimizing GPU inference of a graph transformer for protein docking, in active development. ➤ Implemented LibTorch C++ and TensorRT backends running on an NVIDIA A100. ➤ Profiled with NSight Systems and removed host-device synchronization, reducing forward pass latency by 5x. ➤ Implemented custom Triton kernels for fused scatter-adds, with a 2x speedup over Torch baseline. GPU CUDA Triton PyTorch TensorRT Graph Transformer NSight	2026
GRADINO - OPTIMIZED C AUTOGRAD ENGINE  github.com/SamueleTondelli/gradino Autograd engine written in C optimized for CPU training with SIMD acceleration and custom allocators. C Deep learning SIMD AVX-512 Arena allocators	2026

 LANGUAGES

Italian

English

+ STRENGTHS

- Curiosity
- Motivation
- Passion